Elliptic Modular Forms

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Abstract

Elliptic modular forms are a field of study arising from complex analysis that has proven extremely important to other fields of study. Most notably, the proof of Fermat's last theorem relied on proving a theorem in modular forms called the "Modularity Theorem". In this final project I will provide an introduction to the theory of elliptic modular forms from the perpsective of complex analysis. I will start with the study of elliptic functions and motivate the study of elliptic modular forms in order to expose the deep connections between them. The culminating result in this paper will be a proof of the many named $k/12$ theorem or Valence Formula.

Contents

Introduction

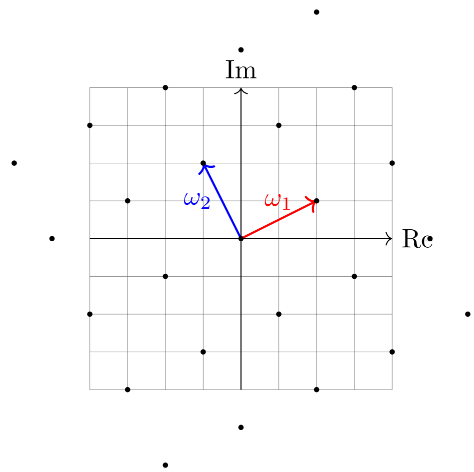
The goal of this paper is to introduce the topic of elliptic modular forms. Readers are expected to be familiar with the standard material of an undergraduate course in complex analysis. I think a lot more exploration of elliptic curves would benefit this paper a lot. As it stands a familiarity with elliptic curves would probably allow a reader to take a bit more from this project. I might try to update this project with that eventually.

Elliptic Functions

Definition 2.1 (Lattice)**.** A lattice in $\mathbb{C}$ is a subset of $\mathbb{C}$ generated by two elements:

$$L = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$$

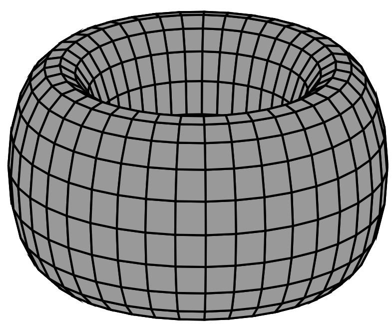
Where $\omega_1, \omega_2 \in \mathbb{C}$ must be $\mathbb{R}$-linearily independant.



Definition 2.2 (Elliptic Function)**.** An elliptic function is a meromorphic function with two independant periods. Explicitly there exists a lattice L such that:

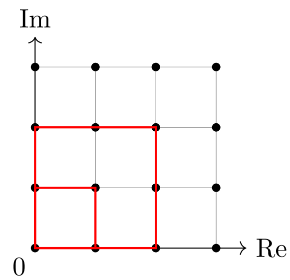
$$f(z + \omega) = f(z) \text{ for all } \omega \in L$$

One may view an elliptic function topologically as a meromorphic function on $\mathbb{C}/L$. This quotient space is topologically equivalent to a torus. This can be seen by glueing a "square" of the lattice to itself where there is periodicity.



This is very important since a torus is compact. Therefore a lattice function may only have finitely many poles and we may assume WLOG for these next few proofs that there are no poles on the boundary of the lattice.

Claim 2.2.1. The integral of an elliptic function around a lattice "square" is 0.



Proof. Suppose that f is elliptic, C is a lattice square and suppose FTSOC:

$$\oint_C f(z)dz \neq 0$$

Then the sum of the residues inside the square is nonzero. Consider:

$$\oint_{2C} f(z)dz$$

By periodicity this line integral should have 2 times the value. However it is around 4 lattice squares and therefore has 4 times the value. This is a contradiction. $\square$

Proposition 2.3 (Liouville Theorems)**.** Any nonconstant elliptic function takes on any value in $\mathbb{C}/L$ the same finite number of times (counting multiplicities). In particular it has the same number of zeros as poles.

Proof. Suppose that f is an elliptic function. Note that the torus $\mathbb{C}/L$ is a compact set. This implies that f has only finitely many poles. Furthermore if f has no poles it attains a maximum modulus on $\mathbb{C}/L$. Then by maximum modulus principle f is constant. We can therefore assume WLOG that f has finite poles. Note that $\frac{f'}{f}$ is also an elliptic function so by the argument principle:

$$\frac{1}{2\pi i}\oint_C \frac{f'(z)}{f(z)}dz = Z - P$$

By Claim 2.2.1 this integral is 0 and f has the same number of zeros as poles. The general result follows by considering the elliptic function $f - z_0$ for any complex number z_0. $\square$

Given an elliptic function f, its order is the number of times it achieves each value in $\mathbb{C}/L$ (counting multiplicities). Note that *Claim* 2.2.1 implies there is no elliptic function of order 1 since it would have a simple pole with nonzero residue.

The Weierstrass $\wp$-function

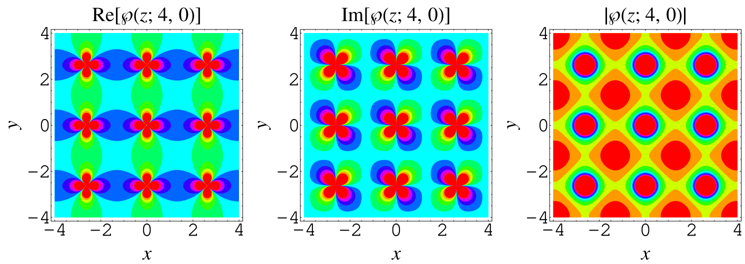
Now that we have some general results, it is important to construct an explicit example of an elliptic function. This can be done by using an infinite sum over the entire lattice to enforce the periodicity requirement. We are furthermore guided by the intuition that this function must have order at least 2.

Example 3.1 (The Weierstrass $\wp$-function)**.** Given a lattice L, the Weierstrass $\wp$-function is an elliptic function defined explicitly by:

$$\wp(z) = \begin{cases} \frac{1}{z^2} + \sum_{\omega \in L\setminus\{0\}} \left[\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\right] & z \notin L \\ \infty & z \in L \end{cases}$$

It's derivative has the form:

$$\wp'(z) = -2\sum_{\omega \in L} \frac{1}{(z-w)^3}$$



The reader is encouraged to convince themselves that this function is indeed elliptic and converges locally uniformly on $\mathbb{C}/L$. This is by a symmetry argument and standard analysis.

Theorem 3.2 (Properties of $\wp$)**.** $\wp$ has the following properties:

(a) It is an even function

(b) It has poles of order 2 exactly on the lattice points.

(c) In $\mathbb{C}/L$ it obtains the values $\wp(e_1), \wp(e_2), \wp(e_3), \infty$ once with multiplicity 2 and every other value twice with multiplicity 1. Here e_1, e_2, e_3 denote that half lattice point $e \notin L, 2e \in L$.

Furthermore $\wp'$ has the following properties:

(a) It is an odd function.

(b) It has poles of order 3 exactly on the lattice points.

(c) It has zeros on exactly the half lattice points e_1, e_2, e_3.

Proof. $\wp 1$, $\wp 2$, $\wp' 1$ and $\wp' 2$ are clear from the formulas for $\wp$ and $\wp'$ respectively (and some manipulation of the sums). Now note that:

$$\begin{aligned} \wp'(e) &= \wp'(e-2e) \\ &= \wp'(-e) \\ &= -\wp(e) \\ \wp'(e) &= 0 \end{aligned}$$

Since $\wp'$ has order three the half lattice points are exactly its zeros with order 1. This proves $\wp' 3$. Now note:

$$z \equiv w \text{ or } z \equiv -w \implies \wp(z) = \wp(w)$$

Therefore every value which is not the image of half lattice point is achieved at least twice with multiplicity 1. By $\wp' 3$ the images of the half lattice points occur with multiplicity. Because $\wp$ has order 2 the implication is double sided. This proves $\wp 3$. □

Definition 3.3 (Eisenstein Series)**.** We define:

$$G_n = \sum_{\omega \in L \setminus \{0\}} w^{-n} \qquad n \in \mathbb{N}, n \geq 3$$

As the Eisenstein series.

If our lattice is of the form $\mathbb{Z}+\mathbb{Z}\tau$ we have the explicit formula:

$$G_k(\tau) = \sum_{0\neq(c,d)\in\mathbb{Z}\times\mathbb{Z}} (c\tau+d)^{-k}$$

This becomes important later.

Theorem 3.4 (Laurent Series of $\wp$)**.** The Laurent series of $\wp$ about 0 has the form:

$$\wp(z) = \frac{1}{z^2} + \sum_{n=1}^{\infty}(2n+1)G_{2(n+1)}z^{2n}$$

Proof. The first term of the Laurent series is obvious, we subtract it and derive. Furthermore since $\wp$ is an even function we can ignore odd order derivatives:

$$\begin{aligned}
\frac{d^{2n}}{dz^{2n}}\left[\wp(z)-\frac{1}{z^2}\right] &= \frac{d^{2n}}{dz^{2n}}\sum_{\omega\in L\setminus\{0\}}\left[\frac{1}{(z-\omega)^2}-\frac{1}{\omega^2}\right] \\
&= (2n+1)!\sum_{\omega\in L\setminus\{0\}}\frac{1}{(z-w)^{2(n+1)}} \\
&= (2n+1)!\sum_{\omega\in L\setminus\{0\}} w^{-2(n+1)} && \text{evaluate at } z=0 \\
&= (2n+1)!G_{2(n+1)}
\end{aligned}$$

Therefore plugging into the Laurent series expansion formula yields the result. □

Note that the elliptic functions on the same lattice form a field under standard sum difference and quotient. We denote $K(L)$ as the field of elliptic functions over the lattice L.

Theorem 3.5 (Structure Theorem)**.** Any elliptic function can be expressed uniquely in terms of the Weierstrass $\wp$-function as:

$$f = R(\wp) + \wp' S(\wp)$$

Where R and S are rational functions. Therefore:

$$K(L) = \mathbb{C}(\wp) + \mathbb{C}(\wp)\wp'$$

Proof. We first prove that any even elliptic function is a rational function in $\wp$. Suppose that f is a nonconstant elliptic function. If f has some pole a not in L we instead consider $h(z) = (\wp(z)-\wp(a))^N f(z)$ where N is the order of the pole. Then h does not have a pole at a. We can therefore assume WLOG that f has poles only in L. Now we induct on n, the order of f. In the base case f is constant and we are done. Otherwise write the Laurent expansions:

$$f(z) = a_{-2n}z^{-2n} + a_{-2(n-1)}z^{-2(n-1)} + \cdots$$

$$\wp(z)^n = z^{-2n} + \cdots$$

We can see that $f - a_{-2n}\wp^n$ has lower order. This completes the inductive step and proves that every even elliptic function is rational in $\wp$.

Now suppose that f is an odd elliptic function. Then $\frac{f}{\wp'}$ is an even elliptic function and can be written as a rational function $R(\wp)$. Then $f = \wp' R(\wp)$ and we are done. Finally suppose that f is any elliptic function. Then f can be split into its even and odd components:

$$f_o(z) = \frac{f(z)-f(-z)}{2}$$

$$f_e(z) = \frac{f(z) + f(-z)}{2}$$

With $f = f_o + f_e$. The result immediately follows. $\square$

The structure theorem essentially states that the Weierstrass $\wp$-function and its derivative are the building blocks for all elliptic functions. This begs the question, given an elliptic function f how can one compute its representation according to Theorem 3.5. The answer is a long and tedious computation involving Laurent series. However sometimes the following relation is helpful:

Example 3.6.

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3$$

Where:

$$g_2 = 60G_4$$

$$g_3 = 140G_6$$

This relation can be verified computationally by bashing out the Laurent series (we only need to check up to the constant term). It can be used to compute the representation of any derivative of $\wp$. Furthermore, it illuminates the connection between elliptic functions and elliptic curves. The Weierstrass form of an elliptic curve

$$y^2 = 4x^3 - g_2x - g_3$$

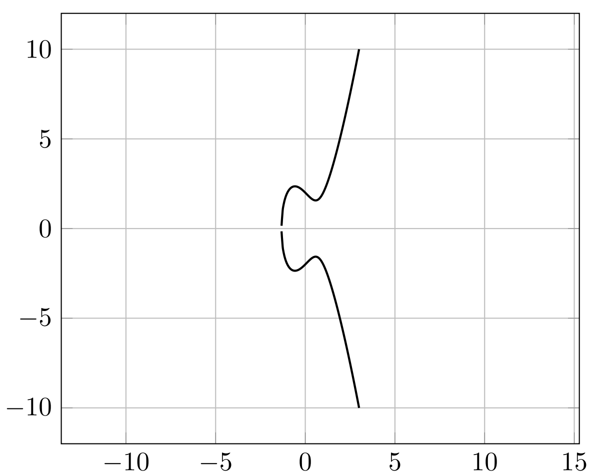
is parameterized by:

$$x = \wp(z) \qquad y = \wp'(z)$$

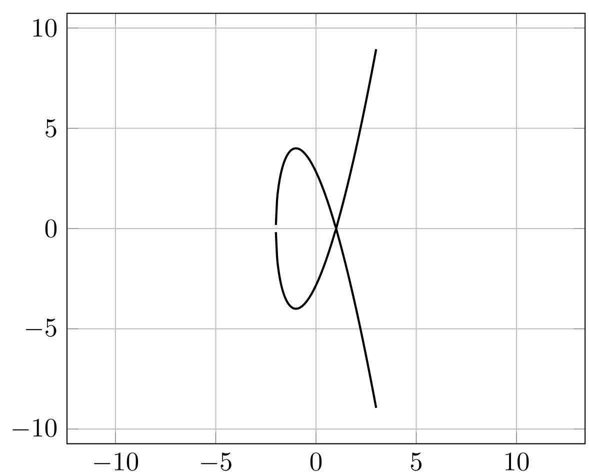
The elliptic curve is nonsingular (has no cusps, self intersections, or isolated points) iff:

$$g_2^3 - 27g_3^2 \neq 0$$

Nonsingular Elliptic Curve: $y^2 = 4x^3 - 4x + 4$



Singular Elliptic Curve: $y^2 = 4x^3 - 12x + 8$



I will not get into the theory of elliptic curves in this paper. I will state one more theorem which I believe provides a lot of insight intro the structure of the elliptic functions. However I will not prove it because the proof requires a lot of results.

Theorem 3.7 (Abel's Theorem)**.** There exists an elliptic function with zeros $a_1, \dots, a_n$ and poles $b_1, \dots, b_n$ iff:

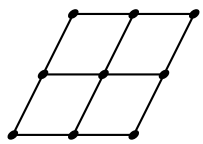
$$a_1 + \cdots + a_n = b_1 + \cdots + b_n \pmod{L}$$

Furthermore this elliptic function is unique up to scaling.

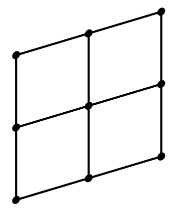
The forward implication is very fun (a clever trick).

More About Lattices

Two lattices are called equivalent if $L_2 = zL_1$ for some $z \in \mathbb{C}$. Using this we can establish an isomorphism between $K(L_1)$ and $K(L_2)$ as $f \mapsto f(z-)$. Therefore elliptic functions over equivalent lattices behave essentially the same.



L_1



$L_2 = zL_1$

We want to investigate when two lattices are equivalent. Note that any lattice is equivalent to a lattice of the form $\mathbb{Z} + \mathbb{Z}\tau$ where Im $\tau > 0$. Therefore we write

$$L_1 = \mathbb{Z} + \mathbb{Z}\tau_1 \qquad L_2 = \mathbb{Z} + \mathbb{Z}\tau_2$$

and claim the following result:

Proposition 4.1. L_1 and L_2 are equivalent iff for suitable integers a, b, c, d.

$$\tau_2 = \frac{a\tau_1 + b}{c\tau_1 + d} \qquad ad - bc = 1$$

Proof. We need $\mathbb{Z} + \mathbb{Z}\tau_2 = z(\mathbb{Z} + \mathbb{Z}\tau_1)$ for some $z \in \mathbb{C}$. Therefore the map $L_1 \to L_2$, $\omega \mapsto z\omega$ is bijective. Surjectivity gives two equations

$$\tau_2 = z(a\tau_1 + b) \qquad 1 = z(c\tau_1 + d)$$

for some integers a, b, c, d. This is written as:

$$z \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \tau_1 \\ 1 \end{bmatrix} = \begin{bmatrix} \tau_2 \\ 1 \end{bmatrix}$$

Injectivity tells use that the inverse of this matrix has integer coefficients, implying that $ad - bc = 1$. Dividing the two equations yields our claim. In the converse direction one can let $z = \frac{1}{c\tau_1 + d}$. ☐

This result motivates us to study Mobius transformations. More specifically what is called the modular group.

Definition 4.2 (Modular Group)**.** The elliptic modular group Γ consists of the transformations of the form:

$$z \mapsto \frac{az + b}{cz + d} \qquad a, b, c, d \in \mathbb{Z} \qquad ad - bc = 1$$

Acting on the upper half plane.

There is an obvious surjection:

$$\mathrm{SL}_2(\mathbb{Z}) \to \Gamma$$

One can show computationally that this is a group homomorphism. Furthermore, two matrices define the same transformation iff they differ by a -1 sign. From now on think of elements of Γ equivalently as "modular matrices" in $\mathrm{SL}_2(\mathbb{Z})$. From the previous exposition there is the following result:

Proposition 4.3. Equivalence classes of lattices are in bijection with $\mathbb{H}/\Gamma$. This is by:

$$\mathbb{Z} + \mathbb{Z}\tau \mapsto \Gamma\tau$$

Recall that elliptic curves of the form $y^2 = 4x^3 - g_2x - g_3$ are paramaterized by $x = \wp$ and $y = \wp'$ for a suitable lattice. Therefore we can understand these elliptic curves by studying the structure of a suitable lattice. This motivates us to import the following quantities from the study of elliptic curves:

Definition 4.4. We have the following quantities:

$$\Delta := g_2^3 - 27g_3^2 \qquad \textit{the discriminant}$$

$$j := \frac{g_2^3}{g_2^3 - 27g_3^2} \qquad \textit{the absolute invariant}$$

With the relations:

$$\Delta(aL) = a^{-12}\Delta(L)$$

$$j(aL) = j(L)$$

For some $a \in \mathbb{C}$.

Proof. Note the following from our exposition on elliptic functions:

$$G_k(aL) = \sum_{\omega \in L\setminus\{0\}} (aw)^{-k} = a^{-k}G_k(L) \qquad g_2 = 60G_4 \qquad g_3 = 140G_6$$

The relations follow by comparing the degrees. □

Recall that Δ tells us whether the elliptic curve $y^2 = 4x^3 - g_2x - g_3$ is singular. Furthermore, j tells us whether two elliptic curves are isomorphic. It is worth noting that $j(\tau)$ is invariant under the action of Γ. We will see that this is important (in particular, j is a modular function). Intuitively, a modular form looks like some sort of lattice invariant. This is the connection between modular forms and elliptic curves. We will eventually prove that j is surjective (so that any elliptic curve has a corresponding isomorphism class of lattices).

Elliptic Modular Forms

Definition 5.1 (Modular Form)**.** A modular function is a meromorphic function with the following property:

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z) \qquad 2k \in \mathbb{Z}$$

We call k the weight of the modular function. A modular form has the additional property of having a non-essential singularity at $i\infty$.

Theorem 5.2 (Modular Group)**.** The modular group is generated by the matrices:

$$T := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \qquad S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Proof. Let Γ_0 be the subset of Γ generated by T and S. First we show any matrix with a 0 entry is in Γ_0. It must have two entries along a diagonal which are ± 1 and one other entry a. We can see that Id, S, S^2, S^3 generate all the possibilities for the diagonal entries. Choose i so that S^i has the desired diagonal entries.

Then either S^iT^a or T^aS^i is the desired matrix (depending on the location of the nonzero entry). Now suppose there is some matrix:

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad M \in \Gamma \setminus \Gamma_0$$

Choose M so that $\min(|a|, |b|, |c|, |d|)$ is minimized among matrices of this type. Suppose that $|a|$ is this minimal element. Note that $|a| \geq 1$ by our result. However:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} ax - b & a \\ cx - d & c \end{bmatrix}$$

This matrix must be in $\Gamma \setminus \Gamma_0$. But since $|a| \leq |b|$ and $|a| \geq 1$ we can choose x so that $|ax - b| < |a|$. This is a contradiction with minimality. The proof for the other entries is similar (with a slightly different choice of matrix) ☐

Therefore the elliptic modular group is generated by the substitutions:

$$z \mapsto z + 1$$
$$z \mapsto -\frac{1}{z}$$

It is worth noting that the Eisenstein series G_k is a modular function of weight k:

Remark 5.3. The following formula holds:

$$G_k\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^k G_k(\tau)$$

Proof. Recall that we proved the converse of Proposition 4.1 by letting $z = \frac{1}{cz+d}$. The result immediately follows from the formula for G_k:

$$G_k(zL) = \sum_{\omega \in L \setminus \{0\}} (zw)^{-k} = z^{-k} G_k(L)$$

☐

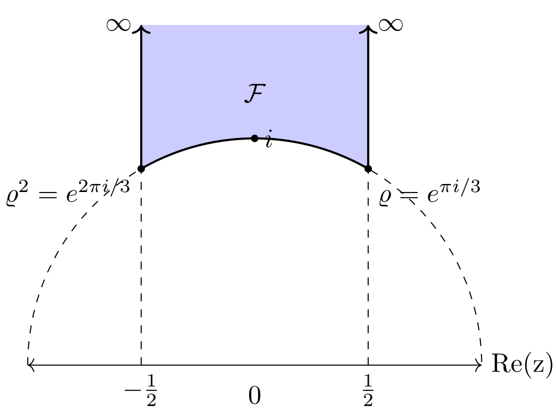
Theorem 5.4 (Fundamental Region of Γ)**.** Consider the "modular figure":

$$\mathcal{F} = \{\tau \in \mathbb{H} : |\tau| \geq 1, |\text{Re } \tau| \leq 1/2\}$$

Then:

$$\mathbb{H} = \bigcup_{M \in \Gamma} M\mathcal{F}$$

And if two points in $\mathcal{F}$ are equivalent mod Γ they must lie on the boundary of $\mathcal{F}$.



Proof. We need to prove for any point $\tau \in \mathbb{H}$ there is a modular substitution $M \in \Gamma$ such that $M\tau$ is contained in $\mathcal{F}$. Let:

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

be a modular substitution and compute:

$$\text{Im } M\tau = \text{Im } \frac{(a\tau + b)(c\overline{\tau} + d)}{(c\tau + d)(c\overline{\tau} + d)} = \frac{(ad - bc)\text{Im } \tau}{|c\tau + d|^2} = \frac{\text{Im } \tau}{|c\tau + d|^2}$$

We can therefore choose M so that the imaginary part of $M\tau$ is maximized. Furthermore the imaginary part of τ does not change if we replace M with:

$$\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} M \qquad \tau \mapsto M\tau + n$$

Therefore we can enforce that $|\text{Re } M\tau| \leq \frac{1}{2}$. Now exploit the maximality condition

$$\text{Im } M\tau \geq \text{Im } \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} M\tau = \frac{\text{Im } M\tau}{|M\tau|^2}$$

to deduce that $|M\tau| \geq 1$ and $M\tau \in \mathcal{F}$ as desired. Since Γ is a group (invertable) this implies that:

$$\mathbb{H} = \bigcup_{M \in \Gamma} M\mathcal{F}$$

Now we prove that this covering is actually a tiling. Suppose that M is a modular matrix so that $M\mathcal{F} \cap \mathcal{F} \neq \emptyset$. Choose some $\tau \in \mathcal{F}$ so that $M\tau \in \mathcal{F}$. Note that $\tau \in \mathcal{F} \implies \text{Im } \tau \geq \frac{\sqrt{3}}{2}$. Then for all integers $\alpha, \beta \neq (0, 0)$ there is the inequality $|\alpha\tau + \beta| \geq 1$. We apply this to τ and $M\tau$ is two ways:

$$|c\tau + d| \geq 1$$

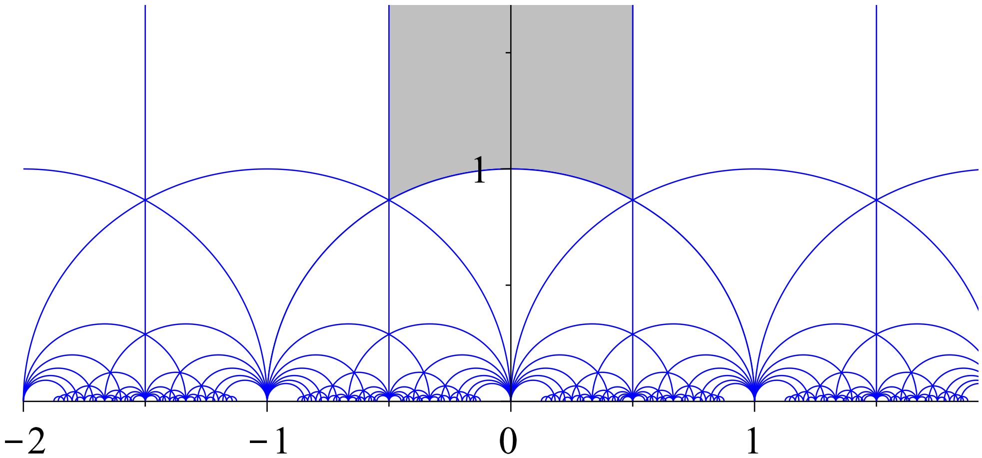
$$1/|c\tau + d| = |-cM\tau + a| \geq 1$$

Therefore $|c\tau + d| = 1$ which implies $c, d \in \{-1, 0, 1\}$. Since $M^{-1}\mathcal{F} \cap \mathcal{F} \neq \emptyset$ we can apply the same argument to yield that $a, b \in \{-1, 0, 1\}$. Now one iterates over all determinant 1 matrices with entries in $\{-1, 0, 1\}$ and computes intersections:

Matrix	Intersection
$\pm \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$\mathcal{F}$
$\pm \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	right vertical boundary
$\pm \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$	right vertical boundary
$\pm \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$	lower boundary arc

In all the other cases there is a one element intersection (either ϱ or ϱ^2). These matrices are displayed in the section on fixed points. ☐

Therefore the images of $\mathcal{F}$ under the transformations in Γ essentially tile the upper half plane. The following image displays this tiling:



Each region is the image of $\mathcal{F}$ under some modular substitution. Our (arbitrarily) chosen fundamental domain is darkened.

Definition 5.5 (Fixed Points)**.** A point $p \in \mathbb{H}$ is called an elliptic fixed point iff the stabilizer:

$$\Gamma_p\{M \in \Gamma : Mp = p\}$$

Contains a nontrivial element. The order of the fixed point is defined as:

$$e(p) := \frac{|\Gamma_p|}{2}$$

The number of modular substitutions fixing p (up to $\pm$ sign).

Theorem 5.6 (Fixed Points)**.** We have:

$$e(a) = \begin{cases} 3 & a \equiv \varrho \pmod{\Gamma} \\ 2 & a \equiv i \pmod{\Gamma} \\ 1 & \text{otherwise} \end{cases}$$

Proof. It is clear the order of a fixed point is invariant under Γ. Therefore we only consider fixed points in $\mathcal{F}$. Of the matrices mentioned in Theorem 5.4 only:

$$\pm\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \pm\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

Have fixed points, these are everything and i respectively. From Theorem 5.4 it suffices to consider modular substitutions which have entries in $\{-1, 0, 1\}$. We want to solve the equation $M\varrho = \varrho$. Bruteforce yields:

$$\pm\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \qquad \pm\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

From the identity $\varrho^2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \varrho$ we see $M\varrho = \varrho \iff \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}^{-1} M \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \varrho^2 = \varrho^2$. This gives the following matrices fixing ϱ^2:

$$\pm\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \qquad \pm\begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} \qquad \pm\begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}$$

$\square$

Valence Formula

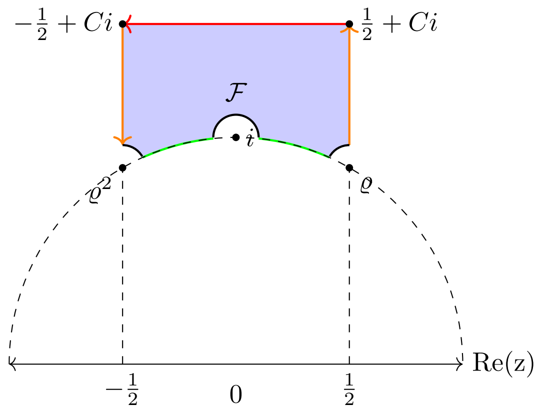
Theorem 6.1 (Valence Formula)**.** If $f \neq 0$ is a modular form of weight k then:

$$\mathrm{ord}(f; i\infty) + \sum_a \frac{\mathrm{ord}(f;a)}{e(a)} = \frac{k}{12}$$

The sum runs over representatives of the poles and zeros of f mod Γ.

Proof. We run over the poles and zeros in the fundamental region $\mathcal{F}$. Since $i\infty$ is non-essential singularity of f we can choose some $C > 0$ so that f has no poles or zeros with imaginary part greater than $\mathbb{C}$. Consider the integral:

$$\frac{1}{2\pi i}\int_{\partial D} \frac{f'(z)}{f(z)}dz \qquad D := \{z \in \mathcal{F}; \mathrm{Im}\ z < C, |z-\varrho|, |z-1|, |z-\varrho^2| > \epsilon\}$$



By the argument principle this integral is:

$$\sum_{a \neq i, \varrho} \mathrm{ord}(f;a)$$

Now we break this integral up to better understand it:

orange

The modular substitution $f(z) = f(z+1)$ tells us the integrals over the orange segments cancel out.

green

Consider the substitution $M : z \mapsto -\frac{1}{z}$ which maps the green segments to eachother (with opposite orientation). We compute:

$$f(-1/z) = z^k f(z)$$
$$f'(-1/z)z^{-2} = z^k f'(z) + kz^{k-1}f(z)$$
$$\frac{f'}{f}(-1/z) = z^2\frac{f'}{f}(z) + kz$$

The one can use a u-sub to compute the integral over the green segments:

$$\begin{aligned}
\frac{1}{2\pi i}\left[\int_{\text{segment 2}}\frac{f'}{f}(z)dz+\int_{\text{segment 2}}\frac{f'}{f}(z)dz\right] &= \frac{1}{2\pi i}\left[\int_{\text{segment 1}}\frac{f'}{f}(z)dz-\int_{\text{segment 1}}\frac{f'}{f}(-1/z)z^{-2}dz\right] \\
&= \frac{1}{2\pi i}\int_{\text{segment 1}}\frac{f'}{f}(z)-\frac{f'}{f}(-1/z)z^{-2}dz \\
&= -\frac{1}{2\pi i}\int_{\text{segment 1}}\frac{k}{z}dz \\
&= -\frac{k}{2\pi i}[\log(\text{end})-\log(\text{start})] \\
&= -\frac{k}{2\pi i}[\log(i)-\log(\varrho^2)] && \text{as } \epsilon\to 0 \\
&= \frac{k}{12}
\end{aligned}$$

red

Noting from the substitution $z\mapsto z+1$ that f and $\frac{f'}{f}$ have period 1. We use Fourier analysis, let $q=e^{2\pi iz}$ and write:

$$\frac{f'}{f}(z)=\sum_{n=0}^{\infty}b_nq^n$$

We can let $C\to\infty$ without changing the value of the integral. Therefore on the red segment all the terms of the Fourier series go to zero except for the constant term b_0. Then the integral over the red line is $-\frac{1}{2\pi i}b_0$. Write:

$$f(z)=\sum a_nq^n$$
$$\frac{f'}{f}=2\pi i\frac{\sum na_nq^n}{\sum a_nq^n}$$

The constant order term in this Fourier expansion is the ratio of the lowest order terms in the numerator and denominator. This can be seen by viewing them as polynomials in q and performing a synthetic division. This ratio just depends on the minimal r so that a_r is nonzero. Then constant term is $b_0=2\pi ir$ and the integral along the red is:

$$-r=-\operatorname{ord}(f;i\infty)$$

black

First we evaluate the black arc around ϱ^2. Consider the power series centered on ϱ^2:

$$\frac{f'}{f}(z)=b_{-1}(z-\varrho^2)^{-1}+\cdots$$

Where $b_{-1}=\operatorname{ord}(f;\varrho^2)$ (this is proven in any proof of the argument principle). As $\epsilon\to 0$ we are essentially integrating $\operatorname{ord}(f;\varrho^2)(z-\varrho^2)^{-1}$ over a small arc of length $\pi/3$. Therefore the integral is:

$$-\frac{1}{6}\operatorname{ord}(f;\varrho^2)$$

By the same argument the integral over the arc around i is:

$$-\frac{1}{2}\operatorname{ord}(f;i)$$

And around ϱ:

$$-\frac{1}{6}\operatorname{ord}(f;\varrho)$$

Note that $\varrho\equiv\varrho^2 \pmod{\Gamma}$ so the integral over the black arcs is:

$$-\frac{1}{3}\operatorname{ord}(f;\varrho)-\frac{1}{2}\operatorname{ord}(f;i)$$

putting it all together

From all these parts we can see:

$$\sum_{a\neq i,\varrho} \operatorname{ord}(f;a) = \frac{k}{12} + \operatorname{ord}(f;i\infty) - \frac{1}{3}\operatorname{ord}(f;\varrho) - \frac{1}{2}\operatorname{ord}(f;i)$$

The valence formula is proven. There is extra work to do if there are poles on the boundary of $\mathcal{F}$, however this can be worked around as seen in [1]. $\square$

Note that Theorem 5.6 makes this sum very easy to calculate.

Applications

I still owe a proof that the j function is surjective:

Theorem 7.1 (j function)**.** The j function is a surjection $\mathcal{F} \to \mathbb{C}$.

Proof. By the open mapping theorem $j(\mathcal{F})$ is open in $\mathbb{C}$. By Theorem 5.4 and the fact that j is Γ invariant $j(\mathcal{F}) = j(\mathbb{H})$. However $\mathcal{F}$ is a compact set in the Riemann sphere $\hat{\mathbb{C}}$. Therefore since j is continuous its image is compact (closed). Since j is nonconstant and $\hat{\mathbb{C}}$ is connected $j(\mathcal{F})$ is the entire complex plane. $\square$

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References

[1] Eberhard Freitag, Rolf Busam, Complex Analysis. Springer